
Fine-Tuning CLIP for Remote Sensing Image Classification

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Abstract

We investigated whether fine-tuning the pretrained vision-language model CLIP can improve zero-shot classification performance on remote sensing imagery. The OpenCLIP ViT-B/32 model was used, and fine-tuned on image-caption pairs from the RSICD dataset using supervised contrastive learning. The models are evaluated using prompt-based zero-shot classification on a separate test split. Results show that training on the dataset substantially improves performance compared to the pre-trained baseline. The best-performing configuration which fine-tunes only the final transformer block of the visual encoder together with the projection layers, achieves a classification accuracy of 73.56% and a macro F1-score of 72.19%, compared to 49.68% accuracy and 42.98% macro F1-score for the pretrained model. The results demonstrate that moderate fine-tuning effectively adapts CLIP to the remote sensing domain while preserving its general visual-language representations.

1 Introduction

Satellite and aerial images are used for tasks ranging from infrastructure analysis, agriculture, and environmental observation. Unlike everyday photos, these images are captured from above rather than from a side view, which can hide lots of details. These images are often quite cluttered and can contain many repeating patterns which makes it even harder to differentiate objects, especially for models pretrained mainly on natural images.

CLIP [1] is a vision-language model that maps images and text into a shared embedding space. Instead of traditional classification with a task-specific classification head, CLIP compares how well an input image’s embedding aligns with the different text embeddings produced from class prompts. This is especially useful as class labels can be represented directly through natural language prompts. While pretrained CLIP has broad knowledge, it is not necessarily optimized for remote sensing imagery.

In this project, we fine-tune OpenCLIP ViT-B/32 on labeled RSICD [2] image-caption pairs and evaluate the model using zero-shot classification. By comparing the pretrained OpenCLIP model with our fine-tuned model under the same evaluation setup, we found that fine-tuning greatly improves classification accuracy and macro F1-score. The fine-tuned model reached a classification accuracy of 73.56% compared to the baseline’s 49.68%, and a macro F1-score of 72.19%.

2 Problem formulation

Let $x_i \in \mathcal{X}$ denote a remote sensing image and let $y_i \in \mathcal{Y} = \{1, \dots, K\}$ denote its land-use class, where $K = 31$ in the RSICD classification setup. The goal is to predict the class label y_i for each image in the test set.

CLIP consists of an image encoder f_{img} and a text encoder f_{text} , which map images and text descriptions into a shared embedding space. For each class $k \in \mathcal{Y}$, we construct a natural language prompt

$$p_k = \text{"a satellite image of } c_k\text{"},$$

where c_k is the name of class k . Given an image x_i zero-shot classification is performed by selecting the class whose text prompt embedding has the highest similarity to the image embedding

$$\hat{y}_i = \arg \max_{k \in \mathcal{Y}} \text{sim}(f_{\text{img}}(x_i), f_{\text{text}}(p_k)).$$

In our experiments, $\text{sim}(\cdot, \cdot)$ is cosine similarity between normalized image and text embeddings.

This project targets the basic version of the assignment. Thus, only Split B of RSICD is used for supervised fine-tuning, using image-caption pairs (x_i, t_{ij}) , where t_{ij} is caption j for image x_i . Split A is not used, and Split C is reserved for final evaluation. The objective of the fine-tuning is to adapt the CLIP embedding space so that images are closer to their matching caption embeddings than to non-matching ones, and without changing the prompt-based zero-shot classification setup.

3 Method

The method follows the basic project version and consists of two parts. Firstly, it is evaluated how well the pretrained CLIP model performs on Split C using zero-shot classification. Secondly, the pretrained CLIP model is fine-tuned on RSICD image-caption pairs from Split B. The same zero-shot evaluation is then repeated on Split C.

3.1 CLIP zero-shot classification

CLIP contains an image encoder f_{img} and a text encoder f_{text} . Both of these encoders will map their input to a shared embedding space, where similarity can be measured using cosine similarity. For zero-shot classification, no separate classification head is trained. Instead each class is represented by a natural language prompt.

For each class name c_k , we construct a prompt

$$p_k = \text{"a satellite image of } c_k\text{"}.$$

All class prompts are then encoded using the CLIP text encoder and L2-normalized

$$z_k^{\text{text}} = \frac{f_{\text{text}}(p_k)}{\|f_{\text{text}}(p_k)\|_2}.$$

Similarly, each test image x_i is encoded and normalized

$$z_i^{\text{img}} = \frac{f_{\text{img}}(x_i)}{\|f_{\text{img}}(x_i)\|_2}.$$

The predicted class is then obtained by selecting the prompt with the highest similarity

$$\hat{y}_i = \arg \max_k \left(z_i^{\text{img}} \right)^\top z_k^{\text{text}}.$$

This procedure is used both before and after fine-tuning. The pretrained model provides a zero-shot baseline, while the fine-tuned model shows whether adapting CLIP to remote sensing captions improves classification performance.

3.2 Supervised fine-tuning

To adapt CLIP to the remote sensing domain, the model is fine-tuned using image-caption pairs from Split B. Each image in RSICD is associated with five captions. These captions are used as text supervision, forming training pairs (x_i, t_i) , where x_i is an image and t_i is one of its corresponding captions.

For a batch of B image-caption pairs, all images and captions are encoded using the CLIP image and text encoders. The resulting embeddings are normalized to unit length. The pairwise similarity matrix is then computed as

$$S_{ij} = \exp(s) \left(z_i^{\text{img}} \right)^\top z_j^{\text{text}},$$

where s is CLIP’s learned logit scale. The diagonal entries S_{ii} correspond to matching image-caption pairs, while off-diagonal entries correspond to non-matching pairs within the same batch.

The training objective encourages matching pairs to have higher similarity than non-matching pairs. The symmetric CLIP contrastive loss

$$\mathcal{L} = \frac{1}{2} (\text{CE}(S, \mathbf{y}) + \text{CE}(S^\top, \mathbf{y})),$$

is used where $\mathbf{y} = (1, \dots, B)$ are the batch indices of the correct image-caption matches. The first term classifies the correct caption for each image, and the second term classifies the correct image for each caption.

After fine-tuning, the model is evaluated using the same zero-shot prompt-based classification procedure as the pretrained baseline. This makes the comparison between pretrained and fine-tuned CLIP direct, since only the encoder parameters have changed while the evaluation protocol remains fixed.

4 Experimental setup

4.1 Dataset

The data used for fine-tuning and evaluating the model is the RSICD dataset, which contains remote sensing satellite imagery of different landmarks and land-use categories such as airports, forests, rivers, residential areas, and playgrounds. Each image is associated with five human-written textual descriptions. Examples of such descriptions include “some building with a parking lot are near an airport with several runways .”, “the forest trees grow very dense .”, and “there are two rows of trees beside the playground .”

The dataset contains a total of 10921 images with a spatial resolution of 256×256 pixels. Both the variety of land-use categories and the multiple captions per image make RSICD work well for fine-tuning and evaluating CLIP in the remote sensing domain.

The original dataset consists of the following files and folders:

- `RSICD_images/`: Contains all satellite images in .jpg format.
- `txtclasses_rsicd/`: Contains multiple text files, where each file corresponds to a specific landmark class. For example, `Airport.txt` contains the filenames of all images representing airports.
- `dataset_rsicd.json`: Contains a list of image filenames together with textual descriptions of the corresponding images in both raw and tokenized format. The tokenized format stores the descriptions as lists of individual words. The file also contains a `split` tag indicating which subset the data belongs to (`train`, `validation`, or `test`).

For convenience during training and validation, the dataset was further processed to simplify handling of the data. First, the dataset was separated into three subsets according to the information in the `split` tag. Data with the `split train` was placed into dataset A (not used in this project), data with the `split validation` was placed into dataset B (used for training the model), and data with the `split test` was placed into dataset C (used for final evaluation).

Furthermore, an additional tag called `label` was added. This tag contains the original class labels obtained from the files in the `txtclasses_rscd/` folder, making the class information directly accessible within the dataset itself. This preprocessing simplified data loading and enabled direct access to class labels during zero-shot classification and evaluation.

During training image-caption pairs from dataset B were used to fine-tune the CLIP model by maximizing similarity between corresponding image and text embeddings. For training, dataset B was further split into a 80-20 training-validation split.

The processed datasets can be found in the `processed_data/` folder, while the script used for preprocessing and dataset conversion is named `split_and_label.py` [3].

4.2 Training details

OpenCLIP [4] with the ViT-B/32 variant from pretrained OpenAI weights was used. The fine-tuning was performed on Split B only. As each image in the RSICD has five captions, each image-caption pair was treated as a separate training example. The images were then processed using the OpenCLIP image preprocessing pipeline, and captions were tokenized using the OpenCLIP tokenizer.

To reduce overfitting on the relatively small labeled training split, we did not update all parameters of CLIP. Instead, partial fine-tuning was performed where different subsets of the model were unfrozen while the remaining parameters were kept fixed. In all cases, the visual and textual projection layers, as well as the logit scaling parameter were kept trainable.

The four main fine-tuning configurations evaluated in this project are:

- **M0:** Only the projection layers and logit scale were trained, while both the visual and text encoders were fully frozen.
- **M1:** In addition to the projection layers, the last transformer block of the visual encoder was unfrozen and fine-tuned.
- **M2:** The last two transformer blocks of the visual encoder were unfrozen, which allows for a larger degree of visual adaptation.
- **M3:** The full visual encoder was unfrozen and fine-tuned, while the text encoder remained frozen.

All models were trained using the symmetric CLIP contrastive loss described in Section 3.2.

We used the AdamW optimizer with a learning rate of 5×10^{-6} for M0 and M1, 3×10^{-6} for M2, and 1×10^{-6} for M3. The batch size was 32, and all models were trained for 15 epochs. The final checkpoint for each configuration was selected based on the highest validation macro F1-score.

4.3 Evaluation

Evaluation was performed on Split C only. Both the pretrained CLIP baseline and the fine-tuned model were evaluated using the same zero-shot classification code and the same prompt template. For each class label c_k , a prompt

$$p_k = \text{"a satellite image of } c_k \text{"}$$

was constructed. All prompts were encoded once with the CLIP text encoder. Each test image was then encoded with the CLIP image encoder, and both image and text embeddings were L2-normalized. Class predictions were obtained by computing the dot product between the image embedding and all class-prompt embeddings, followed by an argmax over classes.

Top-1 classification accuracy is reported as the main metric. For the fine-tuned model macro-F1 is also reported, computed by averaging the F1-score over all classes. Split C was not used during training, hyperparameter selection, or checkpoint selection. A confusion matrix was also plotted to get a better overview over what specific categories the fine-tuned model struggled with.

5 Results

The performance of the pretrained baseline model (B) and the different fine-tuning strategies (M0–M3) is summarized in Figure 1, evaluated using accuracy and macro F1-score on the test set (Split C). The training procedure for all models is described in detail in Section 4.2.

Overall, M1 achieves the highest performance across both metrics, followed by M2, M0, and M3. The pretrained baseline achieved a zero-shot accuracy of 49.68% and a macro F1-score of 42.98%, showing that pretrained CLIP embeddings already contain useful semantic information for remote sensing imagery. However, all fine-tuned models clearly outperform the baseline. In particular, M1 improved the zero-shot accuracy by 23.88 percentage points compared to the pretrained baseline. This shows that supervised adaptation to the remote sensing domain substantially improves the alignment between image and text representations.

The classification done by M1 can be seen in greater detail in the confusion matrix provided in Figure 2. Most of the classes are classified correctly the majority of the time, but there are some specific classes that the model finds very difficult to classify correctly. For example, the class DenseResidential was misclassified as MediumResidential 37 out of 41 times, which is logical because the difference between these classes likely is smaller than the difference between other classes. In contrast, MediumResidential was not misclassified as DenseResidential a single time so M1 appears to have a bias towards classifying samples as "Medium" rather than "Dense". Because the overall accuracy is 73.56% and the errors are concentrated to a small number of class pairs, similar biases likely contribute to the other misclassifications. In general it is clear that most errors are found in the more ambiguous classes, and that the model has higher performance in classes that are not semantically related to other classes.

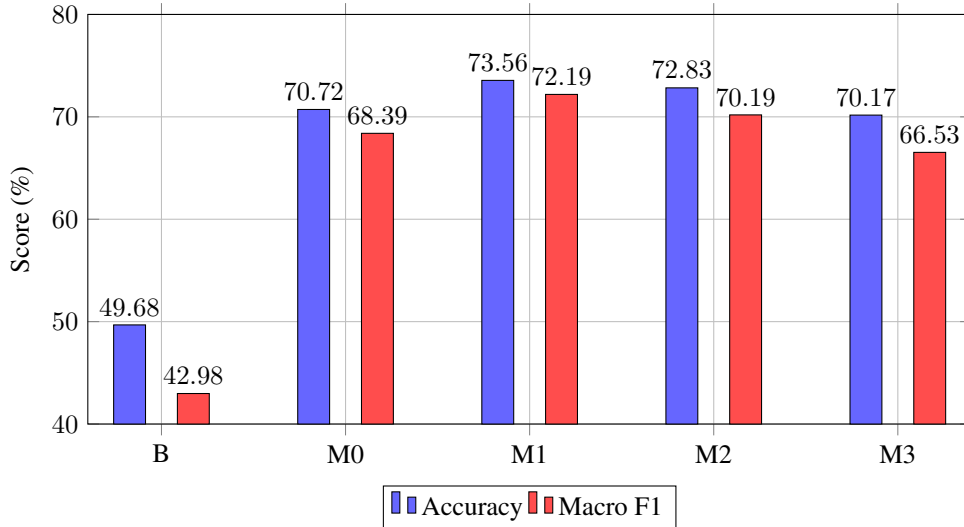


Figure 1: Performance comparison between the pretrained CLIP baseline (B) and different fine-tuning strategies (M0–M3) using varying freezing configurations.

6 Discussion

The results show that supervised fine-tuning of the CLIP model on RSICD image-caption pairs greatly improves zero-shot classification compared to the pretrained CLIP baseline. The pretrained model achieved an accuracy of 49.68%, while the best fine-tuned configuration reached an accuracy of 73.56%. This shows that the original CLIP embedding space already contains useful visual-language knowledge, but by fine-tuning the model with remote sensing captions the performance improves greatly. Other recent works have also reached similar conclusions. Liu et al. [5] continued the training of the baseline CLIP on a large dataset of remote sensing data and found significant zero-shot classification improvements. These results support the conclusion that fine-tuning and adapting CLIP to the remote sensing domain improves performance.

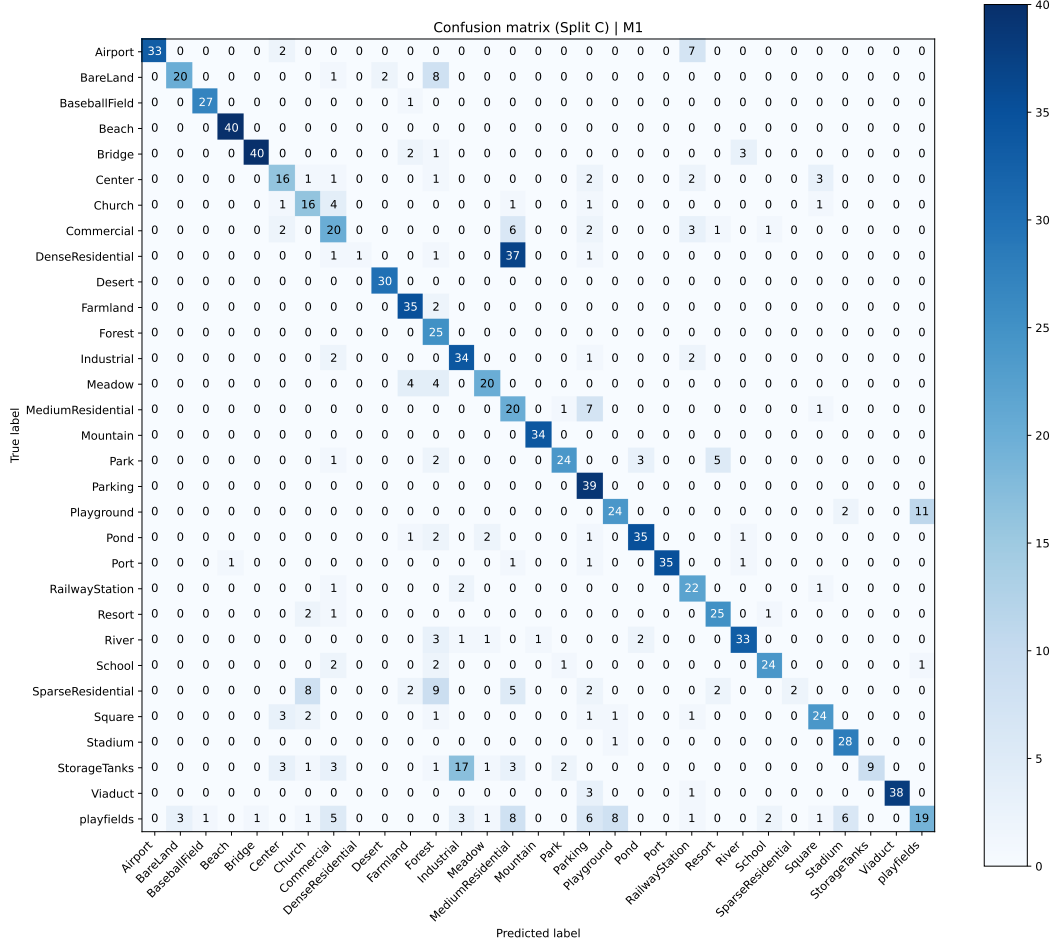


Figure 2: Confusion matrix for zero-shot classification on the test set, using the best fine-tuned configuration (M1).

Among the fine-tuning configurations, M1 achieved the best overall performance. This configuration unfreezes the final transformer block of the visual encoder in addition to the projection layers and logit scale. Compared to M0, which only trains the projection layers, M1 allows the visual representation itself to adapt to remote sensing imagery.

However, unfreezing more of the visual encoder did not further improve performance. M2 which unfreezes the last two visual transformer blocks, performed slightly worse than M1, and M3 which fine-tunes the full visual encoder, performed worse still. A likely explanation is overfitting. Split B is relatively small compared to the size of CLIP, and updating too many pretrained parameters may damage useful general visual-language representations learned during pretraining.

The macro-F1 scores follow a similar trend as the accuracy scores, with M1 obtaining the highest macro-F1. This suggests that the improvement is not only because of better performance on a small number of frequent classes, but also reflects more balanced class-wise behavior.

There are several limitations to the current setup. Firstly, zero-shot classification was performed using only one prompt template, “a satellite image of {class name}”. More robust prompt engineering or averaging over multiple templates could improve performance. Secondly, the basic project setting only uses Split B for training and ignores the unlabeled Split A data, which could potentially be useful for semi-supervised adaptation. More data could potentially allow us to train a more complex model with fewer frozen layers without the risk of overfitting.

Ethical considerations

Remote sensing vision-language models such as CLIP can be used for beneficial purposes such as environmental monitoring, disaster response, and urban planning. However, they may also contribute to large-scale surveillance and military applications, which could be problematic if misused and if privacy is not handled correctly. Another limitation is that CLIP-based models may generate plausible but incorrect predictions, which could result in misinterpretation of satellite imagery. Finally, training and fine-tuning large-scale deep learning models require significant computational resources, leading to increased energy consumption and environmental impact.

Contribution statement

All members in the group contributed to the project as a team. The main part of the technical work like model training, experimentation with freezing strategies, hyperparameter tuning and general coding was done collaboratively mainly over Discord. Filip and Albin took more responsibility for the data preprocessing, such as constructing the labeled splits (A, B, C) and mapping the images to their class labels. Arvid and Victor on the other hand took more responsibility for the code for performance evaluation and analysis of the results. The report was also written collaboratively by all group members, with each group member contributing to multiple sections. How much each group member contributed to each section varied, but every member participated in writing and reviewing the full report as a whole.

AI tools were used during the project only for debugging and coding assistance, questions about report structure, and feedback regarding the report's language. All technical decisions, analysis and method choices were made by the group members without AI assistance.

References

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